Nikhil Maurya

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Professional Summary

Software Engineer specializing in Generative AI and LLM-powered enterprise platforms, combining backend system expertise with hands-on experience in AI agent workflows, prompt engineering, and Model Context Protocol (MCP) integrations. Proven track record in LLM testing, context/memory validation, PII redaction, release engineering, and technical documentation across production AI workspaces, coding assistants, and digital worker automation pipelines.

Technical Skills

Generative AI & Agents: LLMs, Prompt Engineering, AI Agents, MCP (Model Context Protocol), Context & Memory Validation, PII Redaction, Semantic Workflows, RAG Concepts
AI Engineering & Validation: LLM Testing, GenAI Quality Assurance, Prompt Edge-Case Design, Root-Cause Analysis, Release Validation, Defect Triage, Functional & Regression Testing
Languages & Core Tech: Python, JavaScript, Node.js, SQL (PostgreSQL, MySQL), JSON/JSONL, HTML/CSS, React.js, Tailwind CSS, Basic Java, C
Enterprise & Legacy Systems: COBOL, JCL, VSAM, DB2
Developer Tools & Platforms: Git, Azure DevOps, GitHub Copilot, Claude Code, Codex CLI, Gemini CLI, VS Code, Postman, Netlify, Node-RED
Technical Documentation: System Architecture Docs, API/Tool Mappings, Release Sign-Off Reports, User Manuals, Technical Guides

Professional Experience

Cognizant Technology Solutions Aug 2025 – Present
Programmer Analyst – Enterprise AI & Healthcare Solutions *Noida, India*

- Contributed to the end-to-end development, integration, and release validation of enterprise GenAI platforms powering conversational assistants, autonomous digital workers, and code-generation pipelines.
- Architected and executed rigorous test suites for LLM context retention, session persistence, and memory mechanisms across desktop chat and code-generation modules, mitigating context window degradation.
- Integrated and validated Model Context Protocol (MCP) servers and connector configurations, streamlining agentic interactions across Git repositories and Azure DevOps workflows.
- Evaluated multi-agent and digital worker workflows, verifying agent selection accuracy, task routing, human-in-the-loop approval loops, and automated command-loop protection.
- Designed and implemented systematic PII and sensitive data redaction test suites covering credit cards, CVVs, PINs, OTPs, and SSNs to secure inputs and outputs against prompt injection and data leakage.
- Drove branch-level release validation across feature, release, and hotfix builds, performing root-cause analysis across Electron processes, backend service endpoints, and local runtime environments prior to sign-off.
- Authored comprehensive technical documentation, including system architecture overviews, metric-to-tool mappings, user manuals, and structured Markdown deployment reports for engineering stakeholders.

Cognizant Technology Solutions Aug 2024 – Aug 2025
Programmer Analyst Trainee *Chennai, India*

- Supported core backend enterprise workflows involving batch job execution and data processing routines across legacy and relational systems (COBOL, JCL, VSAM, DB2).
- Monitored scheduled batch processing cycles, diagnosed execution abends, and implemented debugging solutions to maintain data integrity and system availability.
- Assisted in defect identification, root-cause debug analysis, and verification across the full software development lifecycle (SDLC).

Cognizant Technology Solutions Dec 2023 – May 2024
GenC Intern *Remote/Hybrid*

- Completed enterprise engineering training covering relational database architecture, batch automation pipelines, COBOL/JCL fundamentals, structured software debugging, and production coding conventions.

Projects

Generative AI Workspace & Intelligent Agent Platform | *Python, LLMs, RAG, AI Agents, MCP, Vector Search*

- Engineered an enterprise GenAI workspace integrating conversational AI, autonomous task planning, multi-step tool execution, and human-in-the-loop approval workflows.
- Implemented end-to-end RAG utilizing document chunking, embeddings, and semantic vector retrieval to enable grounded question answering and dynamic content generation (code, docs, slides, sheets).
- Integrated Model Context Protocol (MCP) specifications to link LLMs with external tools, APIs, and contextual data sources through structured schemas.

- Designed multi-turn conversation memory and session state management to preserve context across long-horizon interactions.
- Constructed automated PII detection and redaction pipelines to sanitize sensitive financial and personal entities (cards, PINs, OTPs, accounts, emails, phone numbers).
- Formulated systematic LLM reliability and edge-case test suites covering context retention, prompt safety, command-loop prevention, tool execution integrity, and failure recovery.

Intelligent Multimodal Biometrics System | *Python, Computer Vision, Machine Learning* | GitHub

- Engineered a high-accuracy multimodal biometric identity verification system integrating palmprint, finger-knuckle, and fingernail modalities.
- Implemented custom feature extraction pipelines and classification algorithms to maximize recognition precision and reduce false-acceptance rates.

Personal Portfolio & Interactive Showcase | *React.js, Tailwind CSS, Netlify* | Live Demo | GitHub

- Designed and deployed a responsive web application featuring custom dark-mode themes, modular components, and automated continuous deployment (CI/CD) pipelines via GitHub Actions and Netlify.

Education

Shri Ramswaroop Memorial College of Engineering & Management 2019 – 2023
Bachelor of Technology in Electronics and Communication Engineering (CGPA: 8.0) *Lucknow, India*

Certifications

AWS Certified Cloud Practitioner – Amazon Web Services
Claude Certified Architect Foundations (CCA-F) – Anthropic
Additional Coursework: Python Programming (Coursera) | Data Analytics with Python (NPTEL) | Generative AI & Prompt Engineering